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习题课13. 6.

设 $f(x) = \ln |p(x)|$, 则 $f'(x) = \frac{p'(x)}{p(x)}$ $f''(x) = \frac{p(x)p''(x) - [p'(x)]^2}{p^2(x)}$

设 $p(x)$ 的零点为 $x_1, x_2, \dots, x_n$, 当 $x = x_i$ 时 ($i=1, 2, \dots, n$), $p(x)=0$, $(p'(x))^2 \geq p(x)p''(x)$

故只需证 $x \neq x_1, x_2, \dots, x_n$ 时, $f''(x) \leq 0$, 即

用 $x_1, \dots, x_n$ 对 R 进行划分, 得到 $(n+1)$ 个开区间, $(-\infty, x_1), (x_1, x_2), \dots, (x_n, +\infty)$, 故只需证 $f(x)$ 在任意上述开区间内是凹函数, 即 $f[\theta t_1 + (1-\theta)t_2] \geq \theta f(t_1) + (1-\theta)f(t_2)$ ($t_1, t_2 \in I$, $\theta \in (0, 1)$)

$$\begin{aligned} \therefore f[\theta t_1 + (1-\theta)t_2] - \theta f(t_1) - (1-\theta)f(t_2) &= \sum_{i=1}^n \ln |\theta t_1 + (1-\theta)t_2 - x_i| - \theta \sum_{i=1}^n \ln |t_1 - x_i| - (1-\theta) \sum_{i=1}^n \ln |t_2 - x_i| \\ &= \sum_{i=1}^n (\ln |\theta t_1 - x_i| + (1-\theta) \ln |t_1 - x_i| - \theta \ln |t_1 - x_i| - (1-\theta) \ln |t_2 - x_i|) \end{aligned}$$

不妨设 $q_i(x) = \ln |x - x_i|$, 易见在任意上述 $(n+1)$ 个开区间内 $q_i(x)$ 是凹函数

$$\therefore f[\theta t_1 + (1-\theta)t_2] - \theta f(t_1) - (1-\theta)f(t_2) = \sum_{i=1}^n (\ln |\theta t_1 + (1-\theta)t_2 - x_i| - \theta \ln |t_1 - x_i| - (1-\theta) \ln |t_2 - x_i|) \geq 0$$

$$\therefore f \text{ 在 } I \text{ 内是凹函数, } \therefore f''(x) \leq 0 \quad \therefore p(x)p''(x) \leq [p'(x)]^2$$